

Carbody Modes and RMS Acceleration

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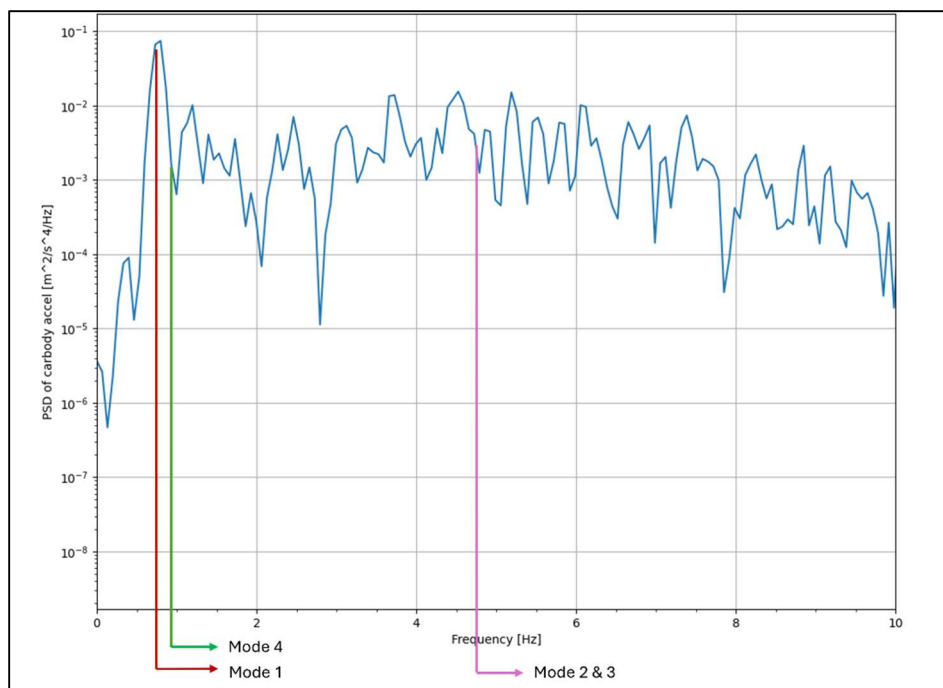
Overview

In response to vertical track disturbances, the railcar body has seven vibrational modes. For the model based on AVE S-103 (ICE3 Velaro E) high speed train, vertical bounce is most strongly affected by frequencies of around 0.75 Hz. Although passengers are most sensitive to vertical disturbances around the 4-8 Hertz range, nausea and carsickness are also experienced due to vertical vibrations below 2 Hertz. Control strategies will focus on reducing vertical bounce at 0.75 Hz, as well as attenuating the 4-8 Hz range.

Vertical Acceleration Data

The figure below presents the Power-Spectral Density (PSD) of the carbody vertical acceleration in response to a rough railroad track input. The railroad is modeled as a stationary ergodic Gaussian process, with irregularities in space created through an inverse fast fourier transform. Vertical acceleration is tracked as it is one of my most significant contributors to passenger discomfort in addition to lateral acceleration (which will not be modeled within the current scope).

Figure 1: Power Spectral Density of Carbody Acceleration from Rough Track Input.



Marked by the red, green, and magenta lines are the natural frequencies of vibrational modes affected by the carbody vertical bounce degree of freedom. Notice the Mode 1 and Mode 4's alignment with the peak at 0.8 Hz. The frequency is most commonly associated with nausea and carsickness. Notice the peak directly prior to the marking for Mode 2 and 3. The peak likely comes before this marking due both to random sampling of the track input, and the fact that resonance generally occurs slightly before natural frequency.

The root mean squared (RMS) vertical acceleration averages to about 0.0092 m/s^2 for six passive control. The RMS suspension travel of the front right bogie (representative of the RMS travel for the other bogies) averages to 0.0004m.

Degrees of Freedom:

1. Carbody Bounce
2. Carbody Roll
3. Carbody Pitch
4. Front Bogie Bounce
5. Front Bogie Roll
6. Rear Bogie Bounce
7. Rear Bogie Roll

Modal Eigenvectors and Eigenvalues:

The table below shows eigenvectors for each mode. The highest value in each column represents that mode's most dominant degree of freedom. Values are unitless; results should only be interpreted relative to one another

Table 1: Each vibrational mode and its associated contributions from each DOF

DOF	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7
1	9.50E-01	2.40E-02	8.74E-18	4.89E-17	0	0	0
2	0	0	0	0	-9.52E-01	-7.65E-03	1.19E-16
3	0	0	-4.75E-03	-3.52E-01	0	0	0
4	2.20E-01	-7.07E-01	7.07E-01	-6.62E-01	0	0	0
5	0	0	0	0	-2.17E-01	7.07E-01	-7.07E-01

6	2.20E-01	-7.07E-01	-7.07E-01	6.62E-01	0	0	0
7	0	0	0	0	-2.17E-01	7.07E-01	7.07E-01
Values	0.75498	4.75471	4.76528	0.94163	0.47907	5.26931	5.26289

Future Steps:

The next section of the project will focus on developing control strategies to reduce the RMS Vertical Acceleration and Suspension travel. Control will be designed to suppress frequencies around 0.8 Hz and attenuate the critical range of 4-8 Hz.

High speed rail cars usually use either active or semi-active controlled suspension. This project will focus on developing a semi-active control system, as the former requires purely actuator-based suspension which is expensive and not viable for the current transportation climate of the United States. Semi-active control generally works by adjusting the viscosity of the fluid within a damper, effectively changing the vehicles damping constants in real time. This allows the suspension system to target certain frequency ranges that need attenuation given the proper sensing and control techniques.